COSC204
Computer Systems

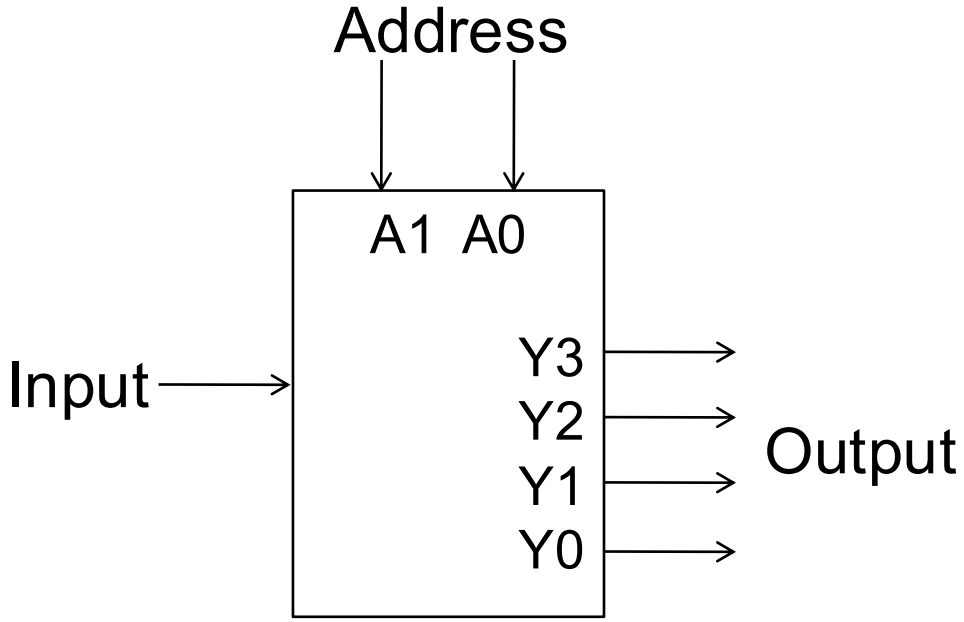
Computer Architecture

Overview

- Last Lecture
  - Latches and flip flops
  - Sequential logic circuits
- This Lecture
  - Computer architecture
  - Sections 3.1 - 3.4, 16.1, and 16.2 of 11th edition
- Next Lecture
  - Machine code
  - Assembly language

Demultiplexor

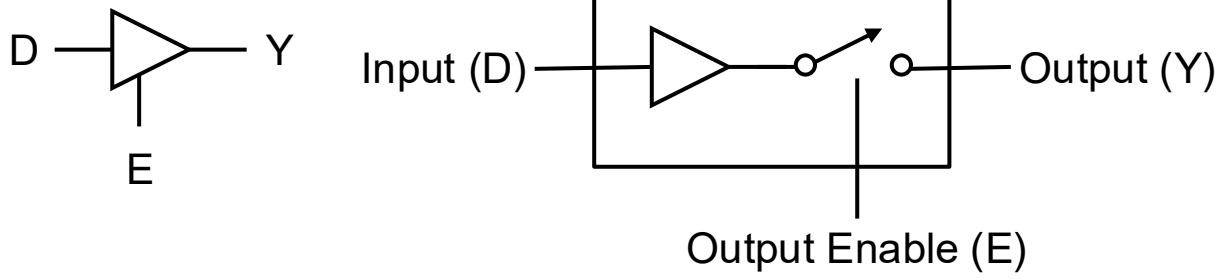
- A demultiplexor is a multi-way switch where the address bits select which output receives the input
- In this example, a 2-bit address selects which of the 4 outputs is connected to the input



A1	A0	Output Location
0	0	Y0
0	1	Y1
1	0	Y2
1	1	Y3

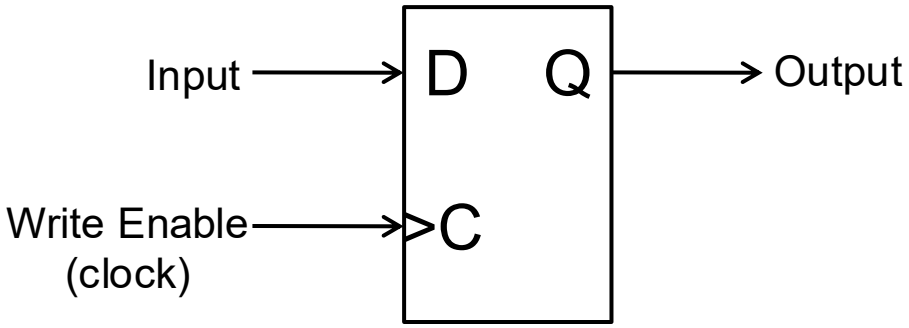
Tristate Buffer

- Three states:
  - High, Low, high-impedance
    - The output is either the input (D) or high-impedance (Z)

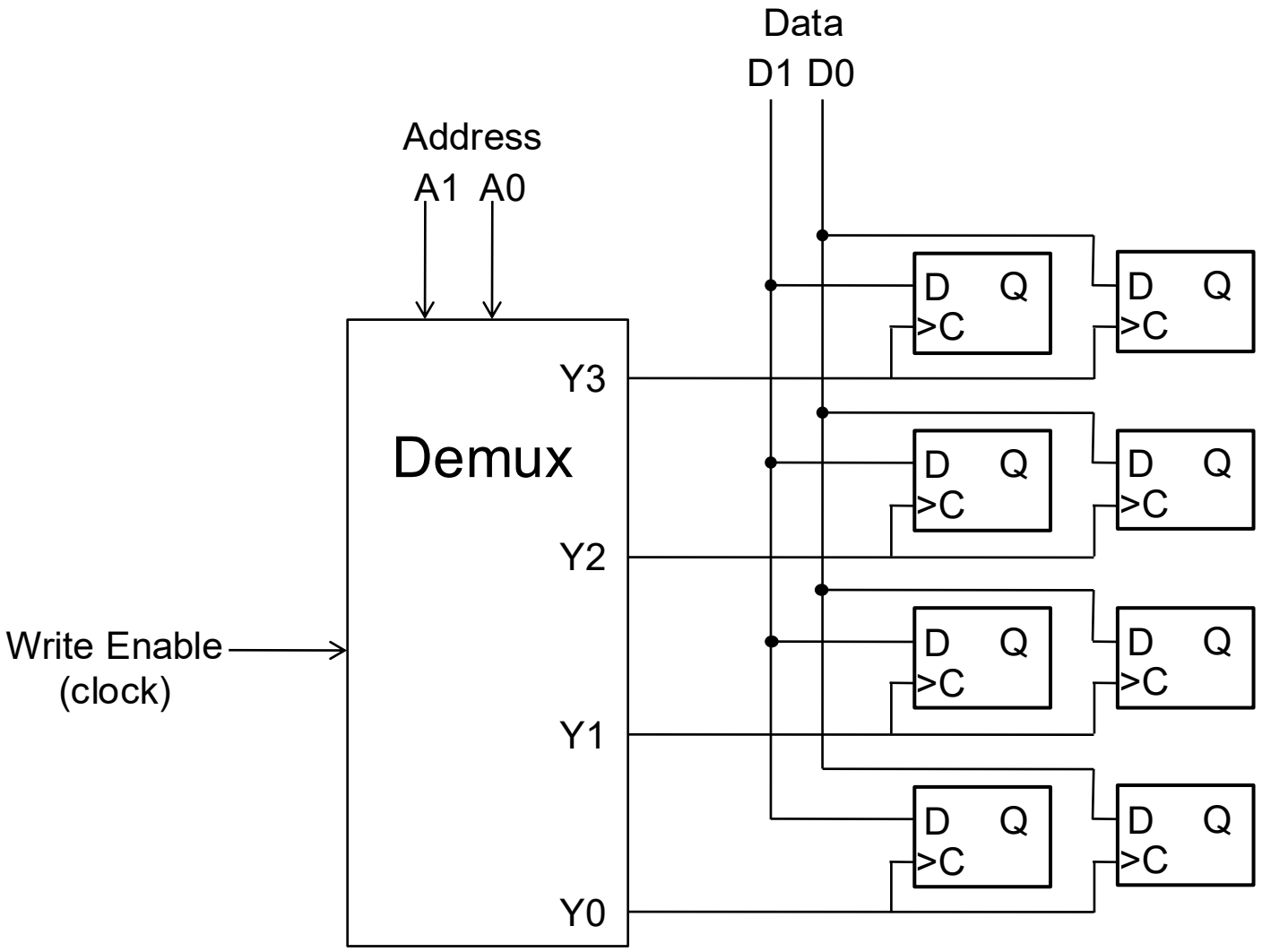


D	E	Y
0	0	Z
0	1	0
1	0	Z
1	1	1

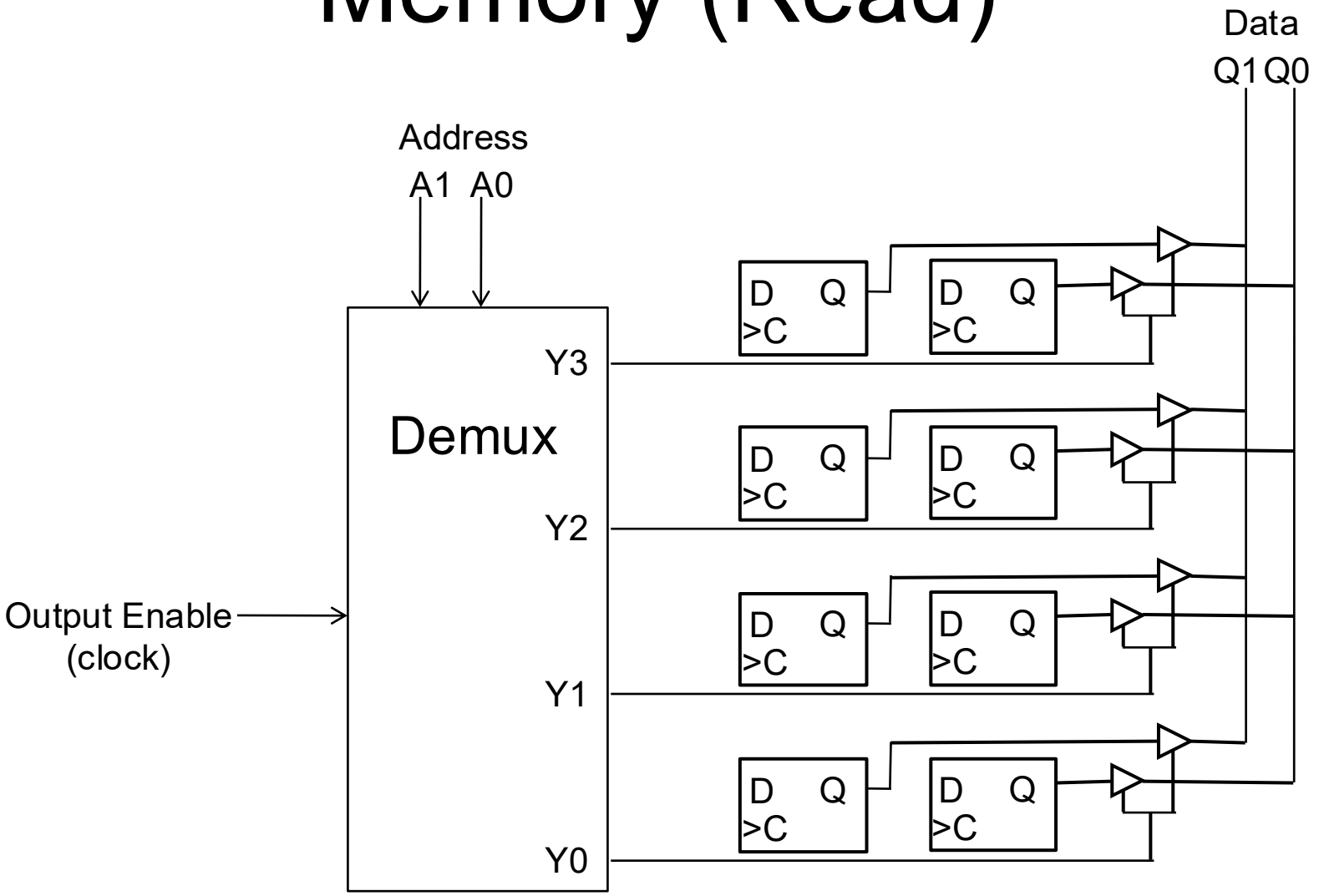
D-type Flip Flop



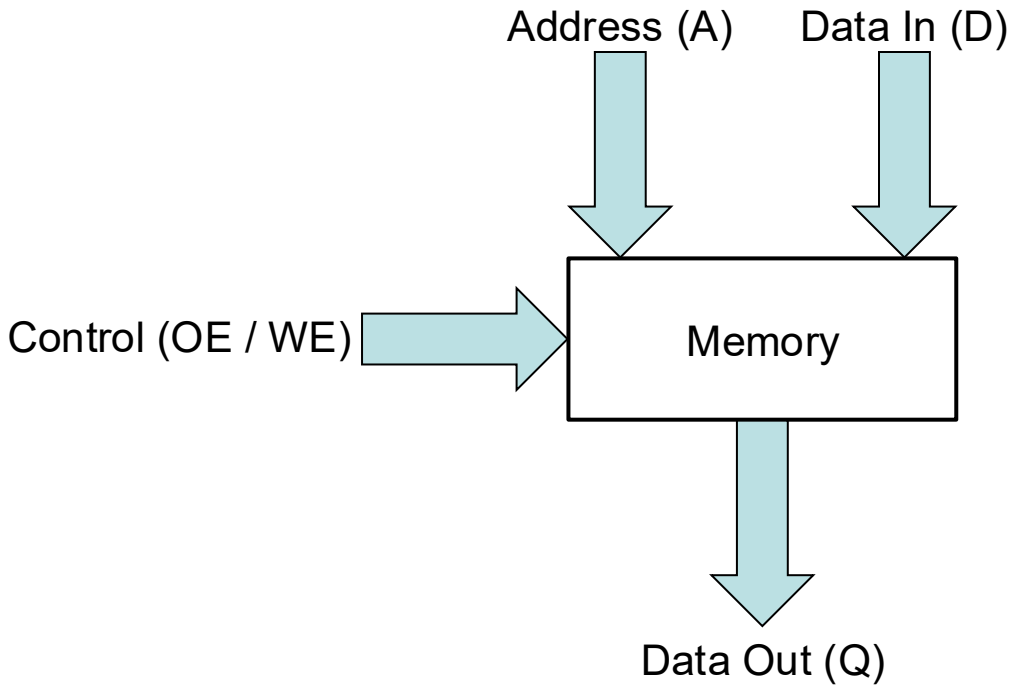
Memory (Write)



Memory (Read)


Data
Q1 Q0
Address
A1 A0
Demux
Y3
Y2
Y1
Y0
Output Enable
(clock)
D Q
>C

Memory (Read / Write)



Bus

- The wide arrows represent several wires
  - We're going to call these arrows "buses"

- The computer has:
  - A data bus
    - D and Q (previous slide)
      - Same wires used for D and Q but never for both at the same time (the bus is "bi-directional"—D is for input, Q is for output)
  - An address bus
    - Which memory cells to use
  - A control bus
    - Output enable and write enable (previous slide) + others

Bus

- Used to communicate between parts of the computer
- Only one transmitter at a time
- Only the addressed device can respond
  - Because the tristate buffers disconnect the other devices
- 2 levels of busses
  - Internal to chip
  - External between chips

Memory

- Memory size is dependant on address bus "width"
  - 2^n bytes for n-lines on the address bus
    - 2^8=256B, 2^{16}=64KB, 2^{32}=4GB, 2^{64}=16EB

Memory

- Flip flops are grouped into bytes (or larger)
  - Each byte has a unique address
  - To write
    - Present the address and the data
    - Tell chip to write
  - To read
    - Present the address
    - Tell the chip to read
    - Look at the data

Static RAM (SRAM)

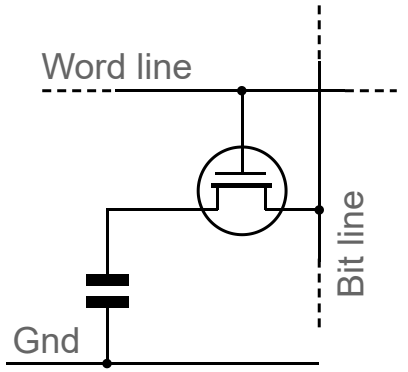
- Memory made from flip flops is called static RAM
  - Used in the CPU cache RAM
  - Used anywhere a small amount of fast RAM is needed
  - But SRAM is expensive

- The main memory of your computer is Dynamic RAM
  - Made from capacitors (little batteries)
  - Used where a large amount of RAM is needed
  - Slower than SRAM
  - DRAM is inexpensive

Dynamic RAM (DRAM)

- Value stored as a charge on a capacitor

  0 = no charge, 1 = charge
- Read from DRAM: write to the *word-line* and read from the *bit-line*
- Write to DRAM: write to the *bit-line* then write to the *word-line*
- Capacitors are *leaky*
  - Need to be periodically refreshed
  - A read operation drains the capacitor

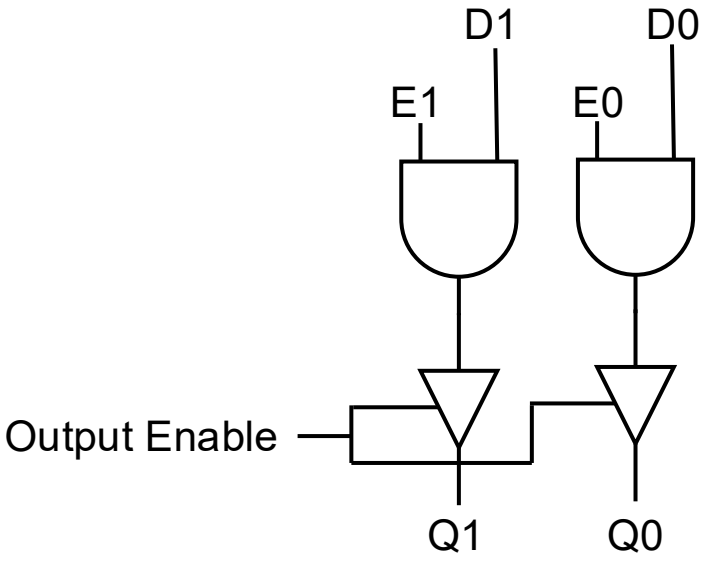


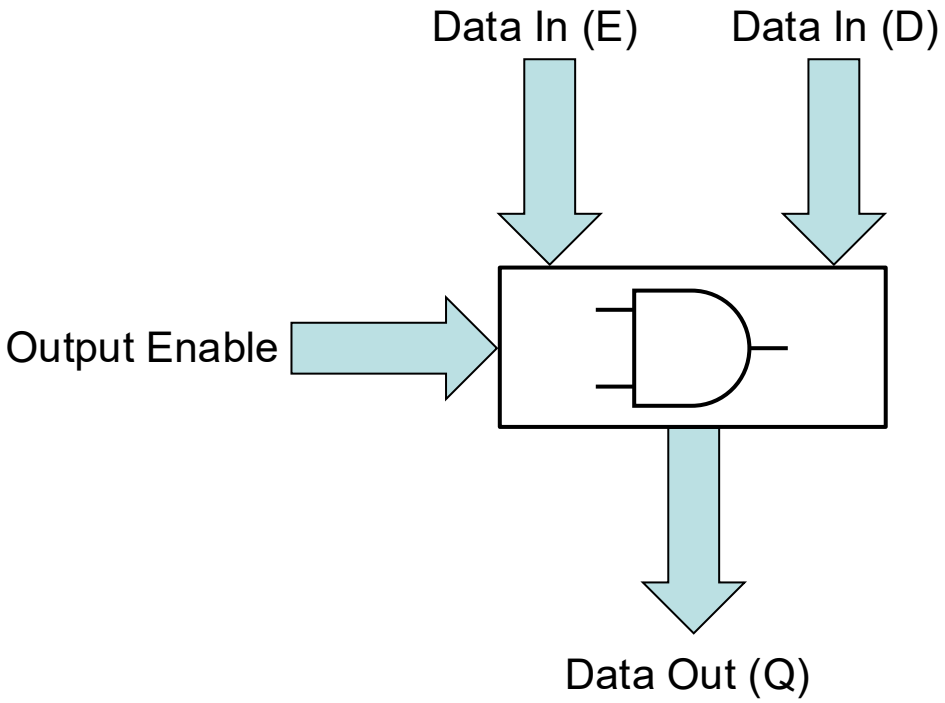
https://www.electronics-notes.com/articles/electronic_components/semiconductor-ic-memory/dynamic-ram-how-does-dram-work-operation.php

Non-Volatile Memory

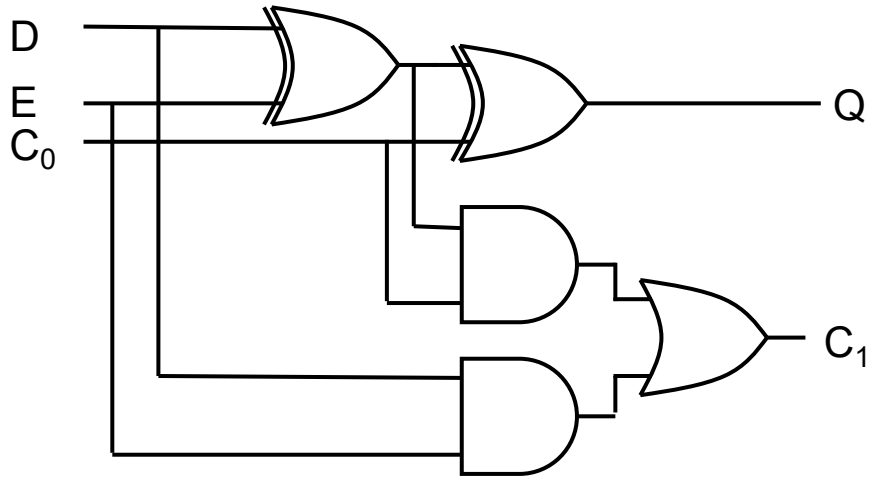
- Often called ROM (Read Only Memory) but may be
  - PROM, EPROM, EEPROM, or FLASH
- FLASH
  - Uses floating-gate flash cells, not quite a transistor
    - https://www.cactus-tech.com/resources/blog/details/solid-state-drive-primer-1-the-basic-nand-flash-cell/

AND architecture package





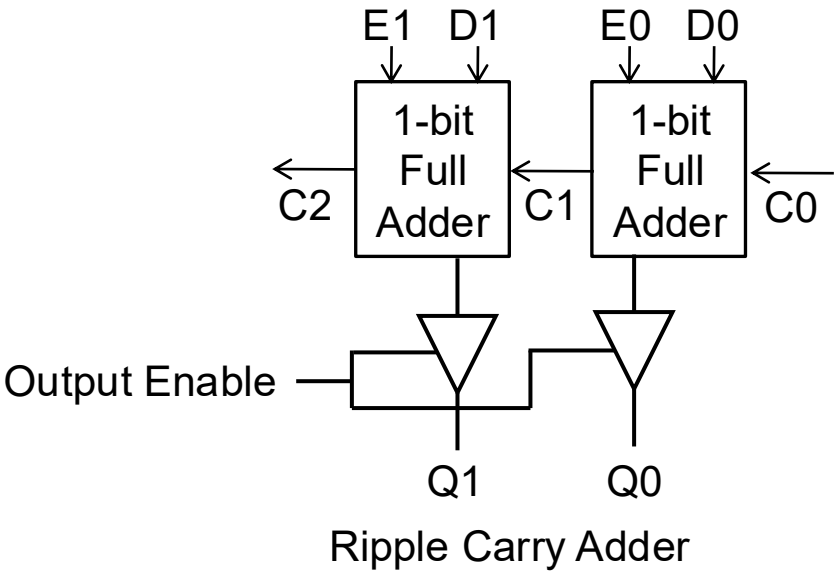
ADD (Plus)

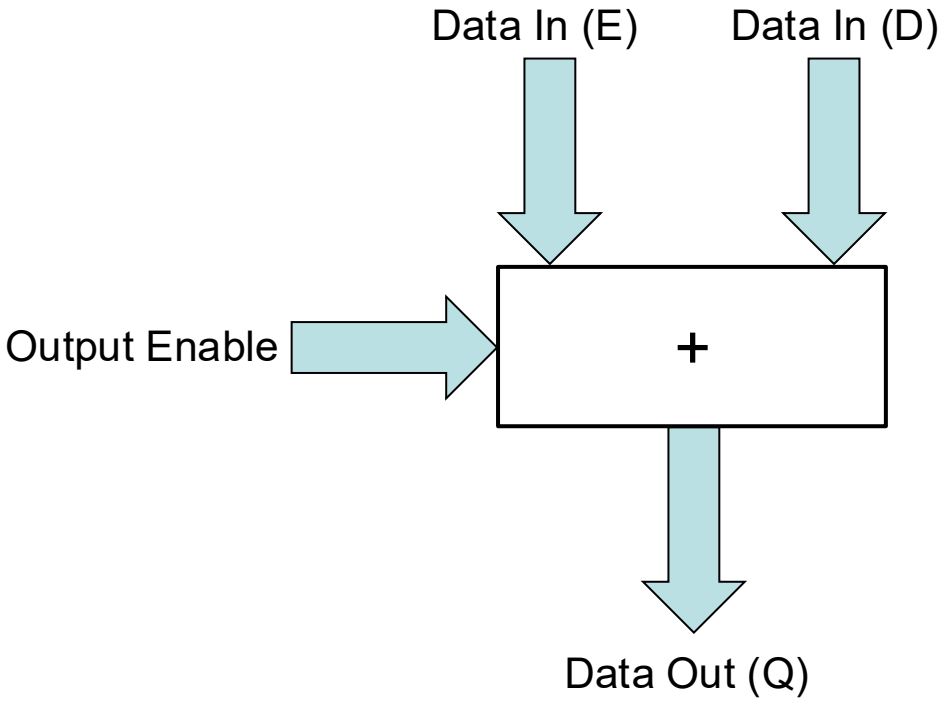


Full Adder

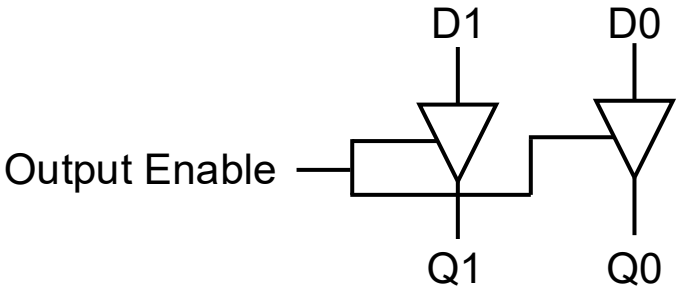
Input			Output	
D	*E*	C_0	*Q*	C_1
0	0	0	0	0
1	0	0	1	0
0	1	0	1	0
1	1	0	0	1
0	0	1	1	0
1	0	1	0	1
0	1	1	0	1
1	1	1	1	1

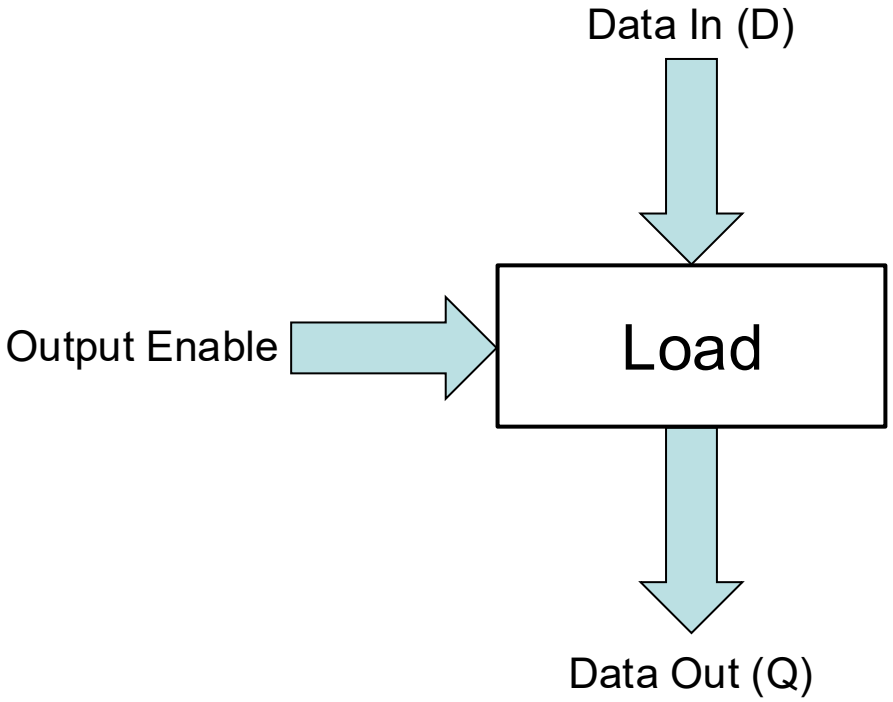
ADD (Plus) – another package



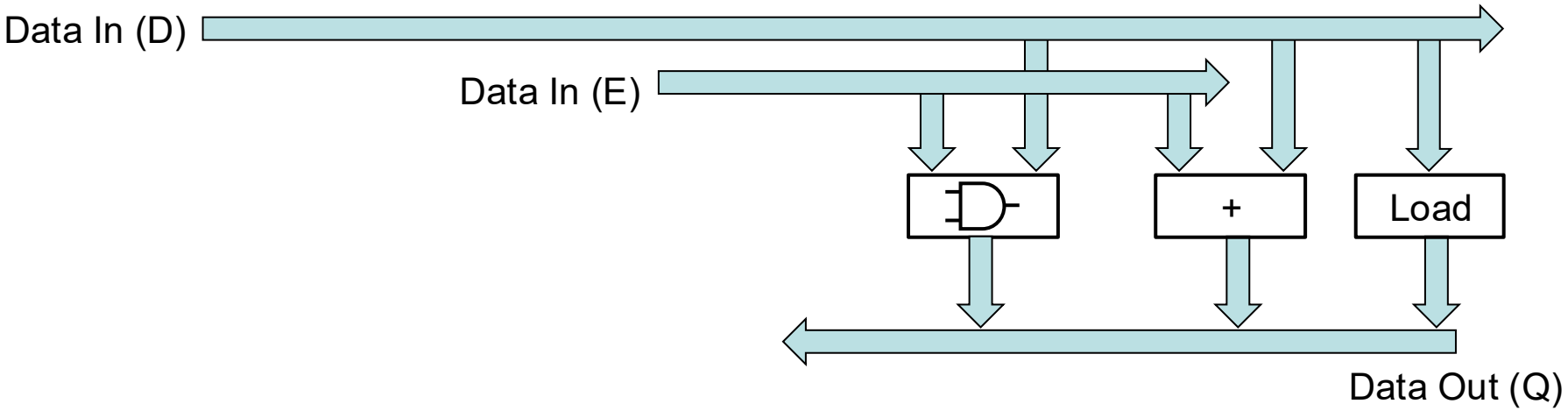


Load


D1
D0
Output Enable
Q1
Q0


Data In (D)
Output Enable
Load
Data Out (Q)

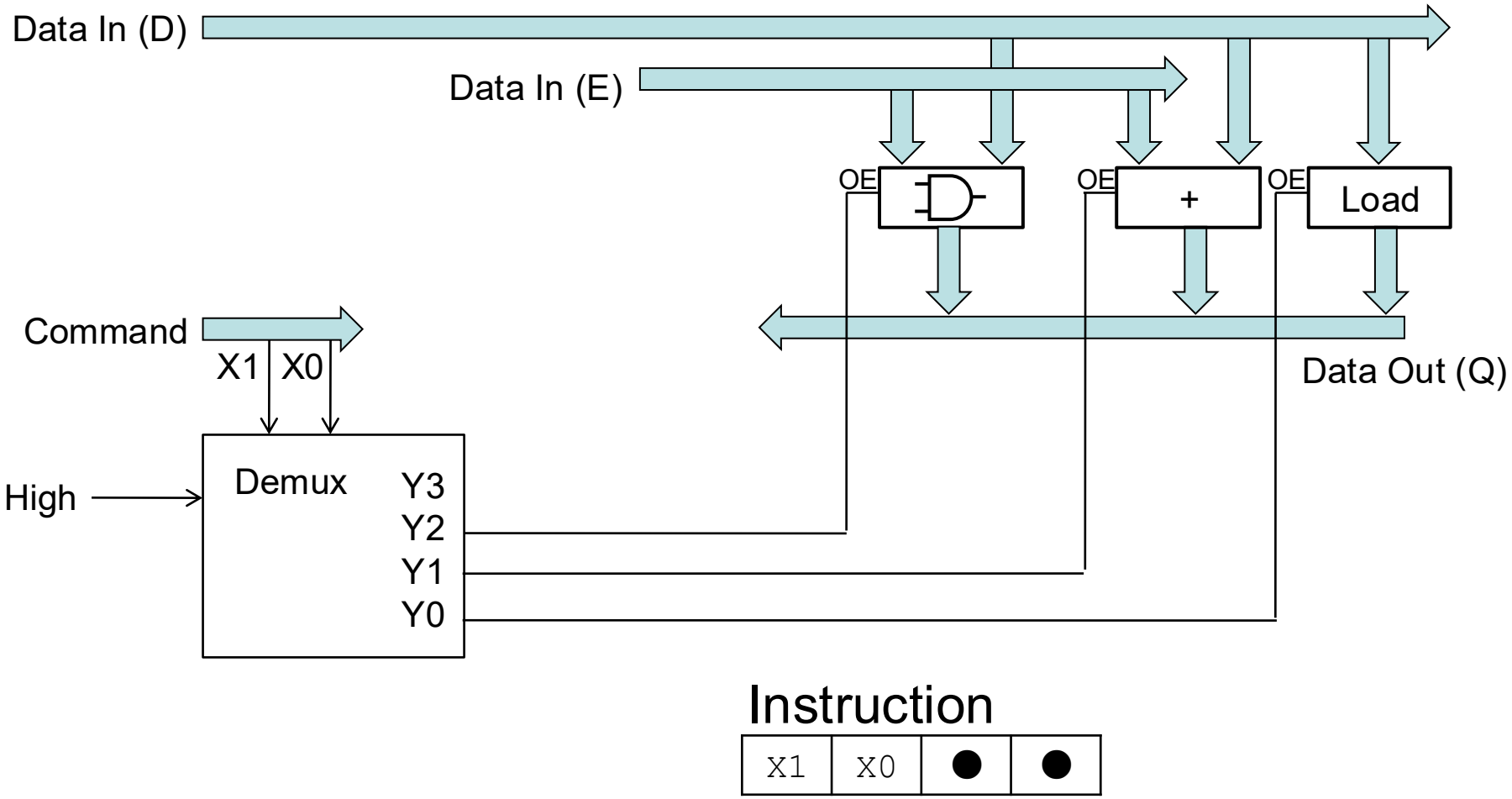
Arithmetic Logic Unit (ALU)



Arithmetic Logic Unit (ALU)

- Performs arithmetic operations
  - addition, subtraction, etc.
- Performs bit-wise logical operations
  - AND, OR, etc.
- Subunits
  - Logical operations (AND, OR, XOR, etc.)
  - Addition (and subtraction using 2's complement)
  - Multiplication and division, if supported
  - Shifting bits left and right
  - Comparison (done by subtraction)
  - Logical tests (if, >0, <0, =0, <=0, etc.)

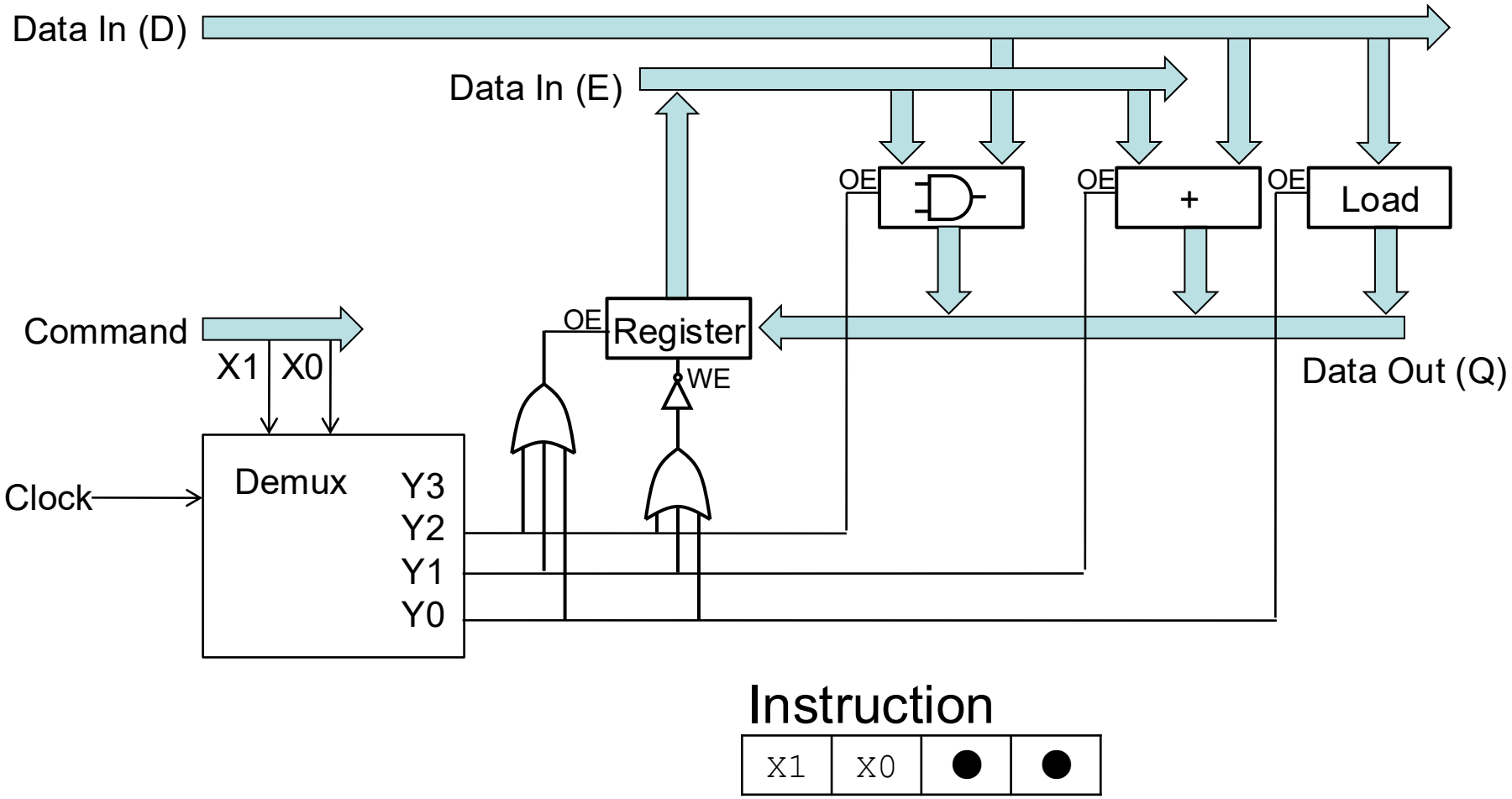
Control Unit



Control Unit

- Coordinates the operation of the computer
- Generates control signals
  - Connects registers to the bus
  - Controls the function of the ALU
  - Provides timing signals to the system
- The actions of the control unit are also associated with the decoding and execution of instructions in a pipelined system

Registers


Data In (D)
Data In (E)
OE
OE
+
OE
Load
Command
X1
X0
OE
Register
WE
Data Out (Q)
Clock
Demux
Y3
Y2
Y1
Y0
Instruction
X1
X0

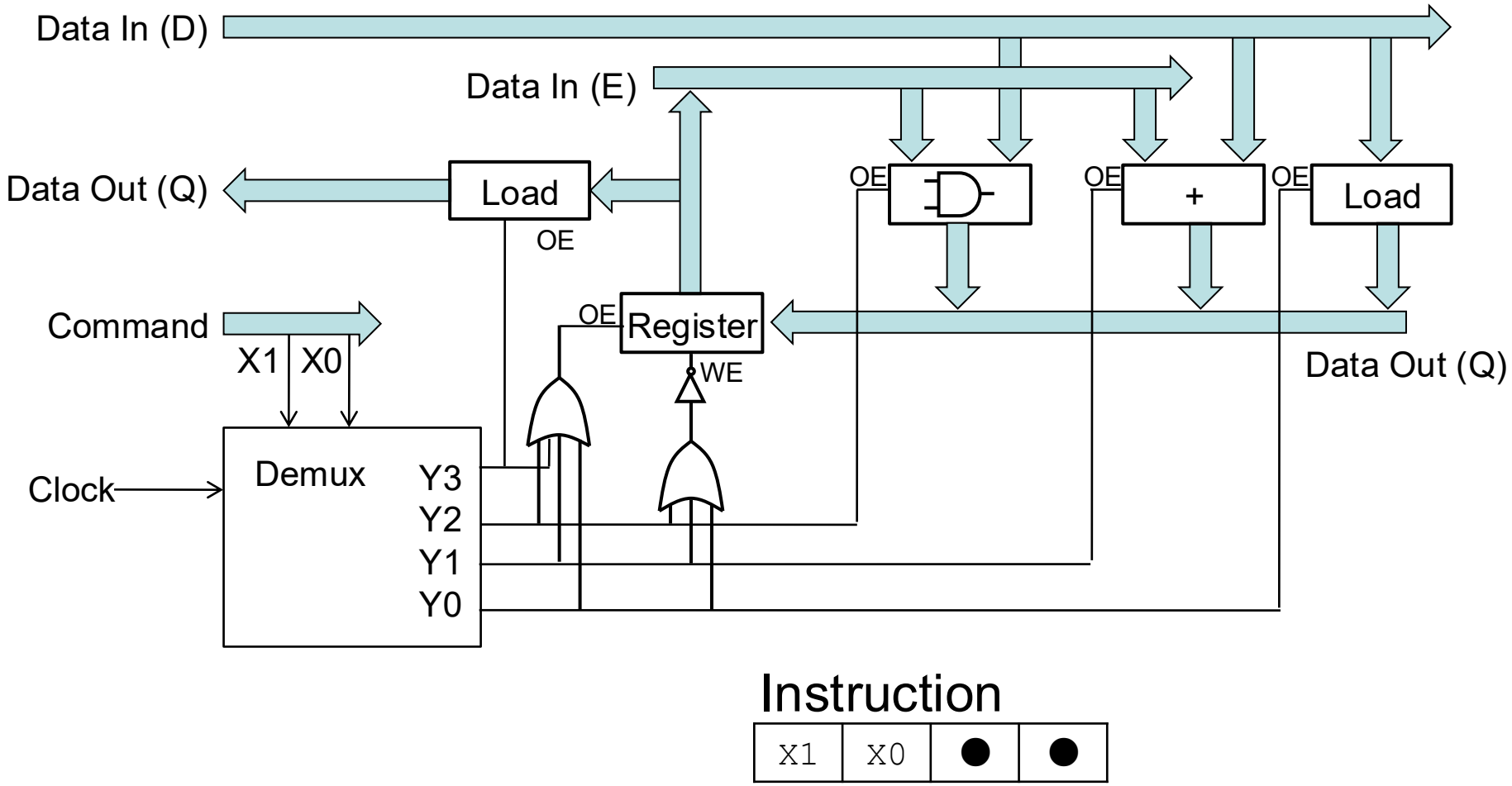
Registers

- Registers are memory cells with names
  - Hold data, or instructions, or CPU status
  - Various sizes (typically: 8, 16, 32, or 64 bits, or larger (512))
  - Registers which hold data typically the same width as memory words
  - Registers for accessing memory typically the same width as the address bus

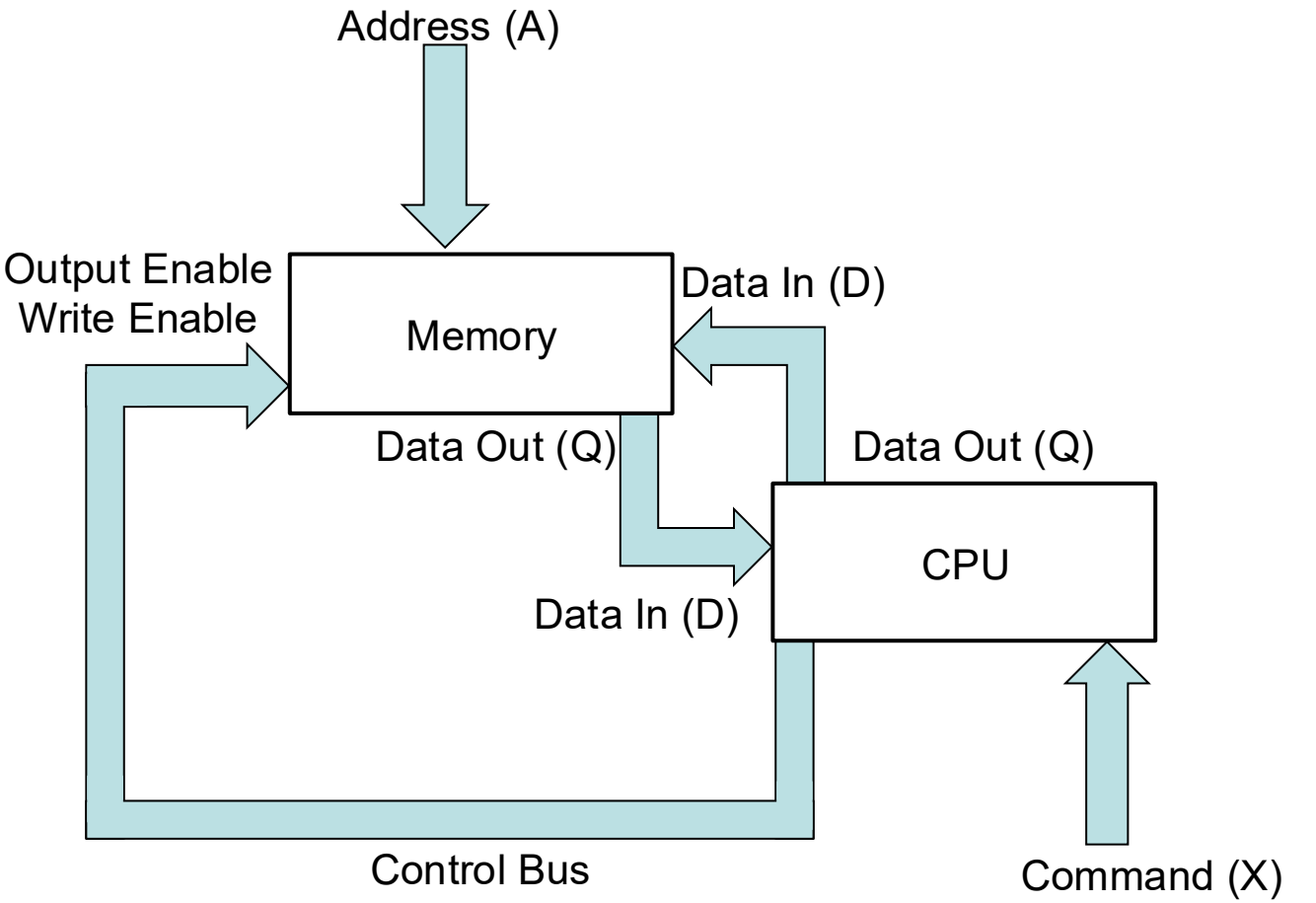
Special Purpose Registers

- The ***Program Counter***
  - Stores the address of the instruction currently being run
- The ***Accumulator***
  - Source of one of the operands
  - Destination of the result in arithmetic operations (plus, etc.)
- The ***Status Flags***
  - Individual bits store information about results of operations
    - Result of last instruction: negative, zero, or positive
    - The carry from the most recent arithmetic operation
    - That overflow occurred during the last instruction

CPU


Data In (D)
Data In (E)
Data Out (Q)
Load
OE
OE
OE
+
OE
Load
Command
X1
X0
OE
Register
WE
Data Out (Q)
Clock
Demux
Y3
Y2
Y1
Y0
Instruction
X1
X0

Computer!



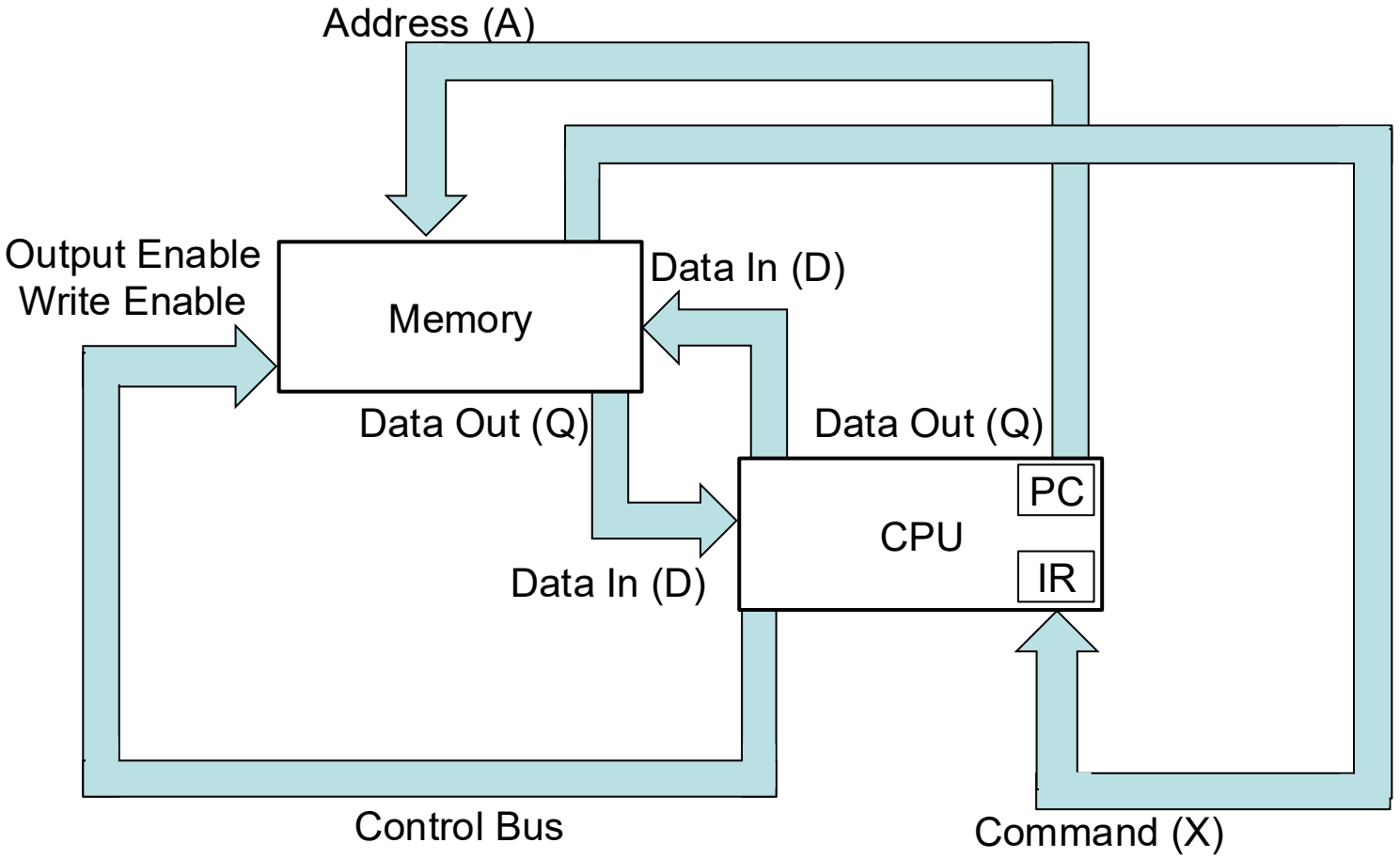
Instruction

X1	X0	A1	A0

Instructions

- Instructions are bit patterns
- Can be split into a number of fields
  - The operation to be executed
  - The address in memory
  - Which registers (or memory cells) to use as operands
  - Where to place the result (registers or memory)

Stored Program Computer



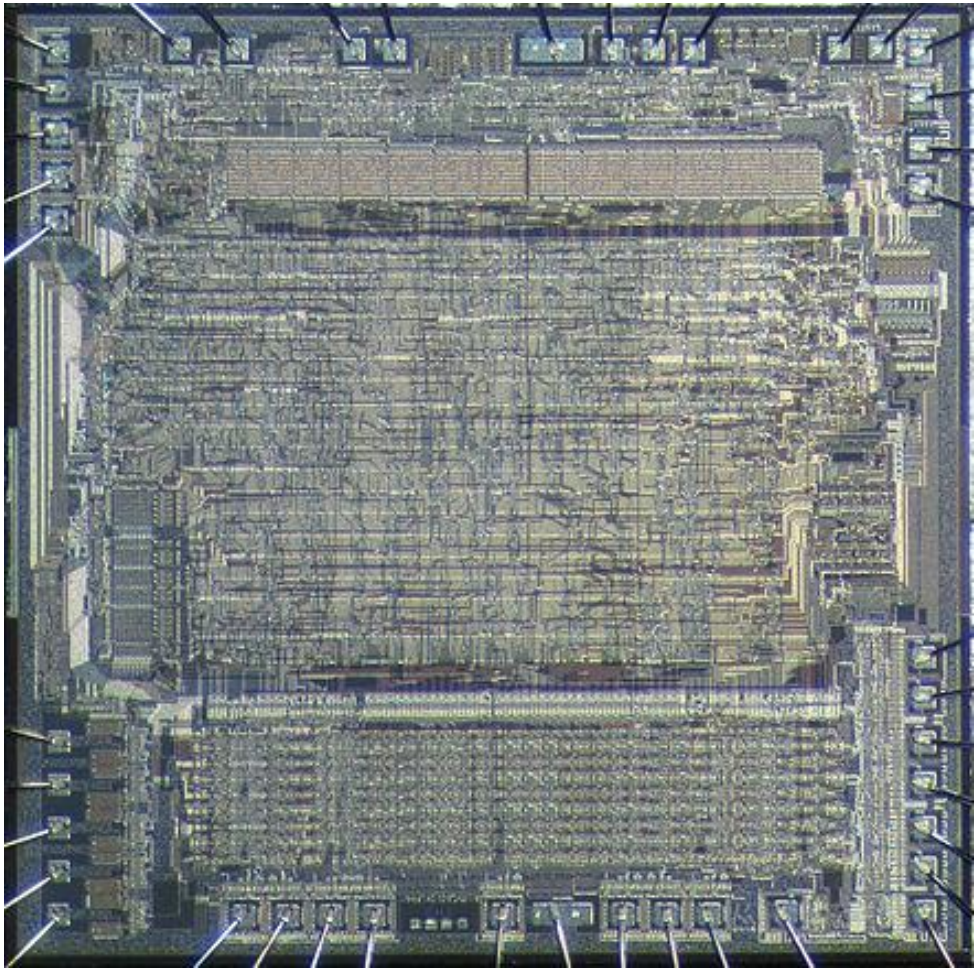
Instruction

X1	X0	A1	A0

Von Neumann Architecture

- Three key concepts:
  - Data and instructions are stored in a single read-write memory
  - Contents of memory are addressable by location, without regard to the type of data contained
  - Execution occurs in a sequential fashion, unless explicitly altered, from one instruction to the next
- Programs and Data are the same thing
  - Therefore, it's possible to write source code (data) and compile them into executables (programs) that can be loaded as data and then executed as programs

Motorola MC6809 CPU



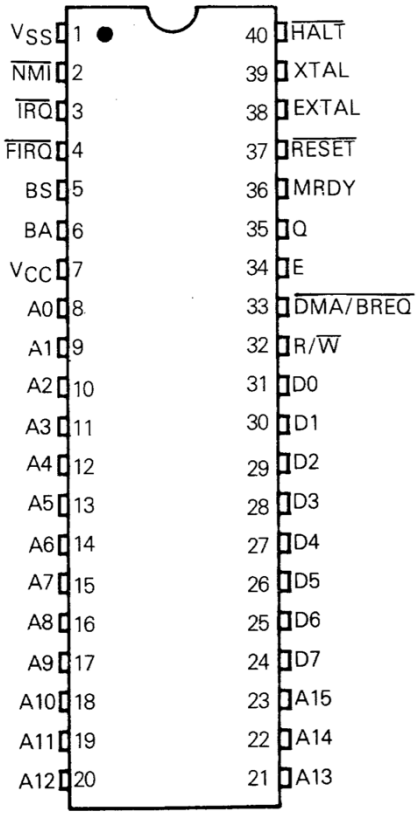
CPU	Year	Transistors
Motorola 6809	1978	9,000
Apple M3 Ultra	2025	184,000,000,000

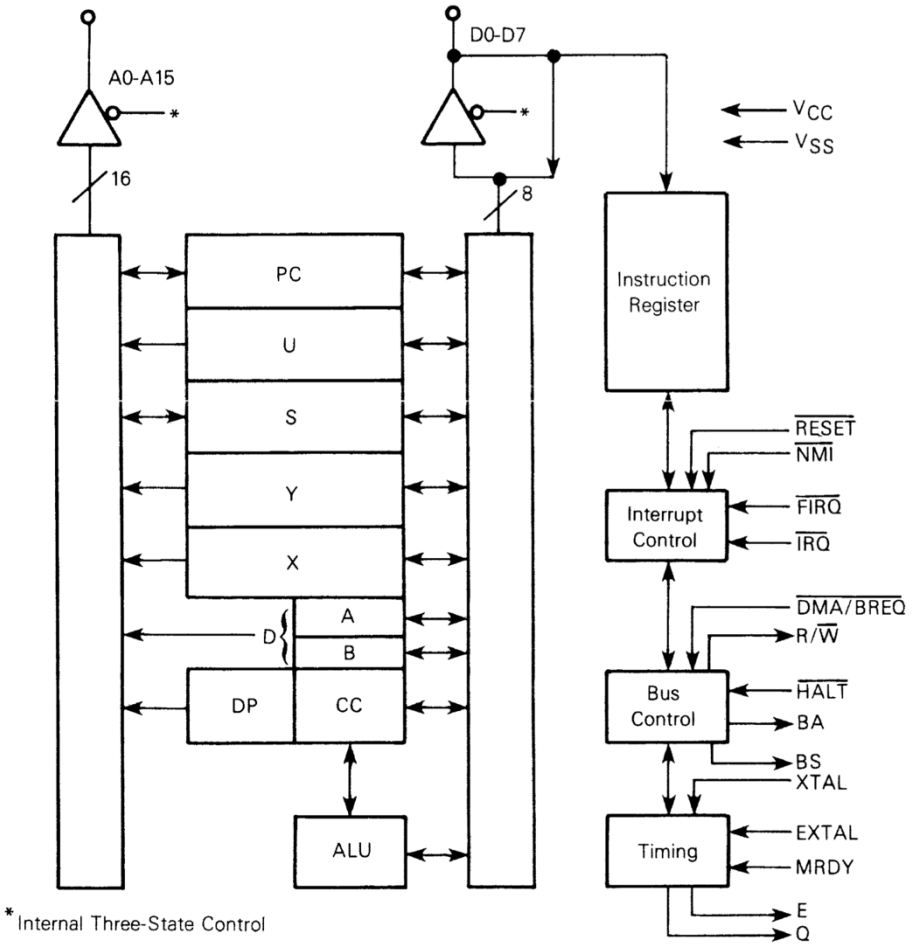
https://en.wikipedia.org/wiki/Transistor_count



https://en.wikipedia.org/wiki/Motorola_6809

Motorola MC6809 CPU





Motorola Semiconductors (1983) MC6809 Datasheet

Homework

- Classify each pin of the 6809 as either
  - Address bus
  - Data bus
  - Control bus
  - Power or ground

- Which pins of the 6809 are connected to memory?
- What is the function on each of the pins in the control bus?